

Divine Nwokedi

CMSC 471

PA1

9/24/2026

- Question 1

- 1) Let n be the current word state and g be the goal state.

$h(n) = \sum(n[i] \neq g[i]), 1 \text{ if true, } 0 \text{ if false}$

It lower bounds the minimum # of single replacements to transform n to g .

To go from g to n , every index where $g \neq n$ requires at least one switch. Since it is uniform cost, this minimum set of switches totals to $h(n)$. Since each letter that doesn't match requires at least one change, the true optimal remaining cost $h^*(n)$ must require AT LEAST that total amount of changes, meaning $h^*(n) \geq h(n)$

- 2) $h(n) = \sum(V(g[i]) \text{ for all } g[i] \neq n[i], V \text{ being the scrabble point value assigned to a letter character.}$

It lower bounds the cumulative insertion costs of the target letters required at the mismatch positions to form g .

In this cost scheme, the minimum cost of a state is the total scrabble points gained of each letter in g that doesn't match the respective letter of n . Some letters will need to be changed multiple times to be able to match g , but at the end, every letter will at some point need to be changed to each letter in g . Therefore, every true optimal remaining cost is forced to be equal to or greater than the scrabble score of each non-matching letter in g .

- 3) $h(n) = \sum(n[i] \neq g[i]), 1 \text{ if true, } 0 \text{ if false}$

It lower bounds the minimum base transition cost of the minimum needed letter replacements.

Entering word w costs $1 + \text{rarity}(w)$, so the minimum cost will be $\text{cost} = 1 + 0.0 = 1.0$. Because every different letter results in a new word with a minimum cost of 1.0, the true optimal remaining cost of a state is $1.0 * k$, given that each future word has a rarity score of 0 and k being the number of non-matching letters

- Question 2

Pair	Cost Measure	UCS Cost	UCS Expanded	A* Cost	A* Expanded	Reduction %
WHY ASK	STEPS	5.00	773	5.00	16	97.9
WHY ASK	SCRABBLE	11.00	765	11.0	13	98.3

WHY ASK	FREQ	38.86	808	38.86	755	6.6
BEAR DUCK	STEPS	4.00	691	4.00	6	99.1
BEAR DUCK	SCRABBLE	11.0	2981	11.00	6	99.8
BEAR DUCK	FREQ	38.24	1084	38.24	720	33.6

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- Both UCS and A* returned the same optimal cost, verifying that the heuristics that were applied are admissible.
- For WHY->ASK: Scrabble gave the greatest reduction; for BEAR->DUCK: Scrabble gave the greatest reduction. For both pairs, the heuristic was the least informative under the frequency cost measure.
- The Scrabble heuristic accumulates character specific weights, and when higher value letters are targeted, h produces larger lower bound estimates early in the search tree, which results in greater $f(n) = g(n) + h(n)$ for paths that are not promising, allowing A* to deprioritize them early. Under the frequency cost metric, rarity values can be very significant values. However, $h(n)$ for frequency cost calculates lower bound using simple mismatch counts, with rarity value equaling 0. This difference between the absolute minimum cost (1+0) and the potential true minimum cost (1+20+) causes a very loose lower bound, and since $h(n)$ contributes very little relative to $g(n)$, A* acts very similarly to UCS, causing minimal reduction
- Question 3
 - Steps: YES; Since every change has a unit cost of one, if an optimal path exists in one direction, it will exist in the other direction as well due to word transformations being symmetrical (ex. Car -> cat is the same cost as cat -> car).
 - Scrabble: NO; Changing a character from c-old to c-new yields a cost of $V(c\text{-}new)$. Meanwhile, changing a character from c-new to c-old (opposite direction) yields a cost of $V(c\text{-}old)$. If these are not equal, the transitions are not symmetrical. For example: dog->cog: $V('c') = 3$; cog->dog: $V('d') = 2$; These are not symmetrical.
 - Frequency: NO; The initial word is the only word that doesn't add to the frequency cost, but all future transition and goal words do. If the initial and goal words do not have the exact same rarity, the total cost of moving to a word is non symmetrical. For example: let x be the rarity of dog and y be the rarity of cog and $x \neq y$; dog->cog: $1+y$; cog->dog: $1+x$; therefore, the cost of the transition is not equal to the cost of its opposite.
- Question 4

- Model: GPT-4o; Proposed heuristic: "Define $h_{\text{LLM}}(n) = h_0(n) + \text{rarity}(g)$ for $n \neq g$, and $h_{\text{LLM}}(g) = 0$, where g is the goal state. Since reaching g requires entering word g at least once at the final step, adding $\text{rarity}(g)$ to $h_0(n)$ lower-bounds the unavoidable cost."
- LLM Proposal: $h(n) = n=g ? 0 : h_0(n) + \text{rarity}(g)$
- An admissible heuristic would require $h(g) = 0$ at the goal state. This is where the LLM proposal could potentially lose admissibility: it depends on how the goal state is handled. At the goal state, $h(g) = 0$ must hold true, since if the heuristic isn't set directly for $h(g)$, it would result in $h(g) = h_0(n) + \text{rarity}(g)$, and would break admissibility provided that $\text{rarity}(g) > 0$. However, since the LLM defined a piecewise check, it protects the heuristic from this edge case, and since this was the only case where admissibility is challenged, the proposed heuristic holds admissible.